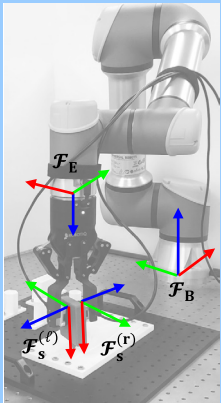
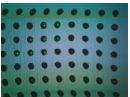
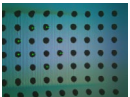


Data Acquisition



RGB tactile images


 $\mathbf{I}_t^{(\ell)}$

 $\mathbf{I}_t^{(r)}$

End-effector pose

$${}^B\mathbf{T}_E(t) \in SE(3)$$

Sensor-to-EE transforms

$${}^E\mathbf{T}_S^{(\ell)}, {}^E\mathbf{T}_S^{(r)}$$

Motion Signal Extraction

3D marker tracking

- Optical flow tracking + Depth reconstruction

$$\mathbf{x}_{i,t}^{(\ell)}, \mathbf{x}_{i,t}^{(r)} \in \mathbb{R}^3$$

Coordinate transformation

- Sensor $\rightarrow$ EE $\rightarrow$ World

$$\mathbf{X}_t \in \mathbb{R}^{2K_m \times 3}$$

6D tool motion

- Kabsch alignment (initial $\leftrightarrow$ current)

$$(\mathbf{R}_t, \mathbf{d}_t)$$

Two-Stage Contact Inference

Stage I: Initial contact anchoring

- time $t = 0, 1, \dots, \tau_{cal}$
- Fixed-point constraint

Global initial contact location

$$\hat{\mathbf{p}}_B(0) = \arg \min_{\mathbf{p}} \sum_t \|(\mathbf{I}_3 - \mathbf{R}_t)\mathbf{p} - \mathbf{d}_t\|_2^2$$

Stage II: Trajectory reconstruction

- time $t = \tau_{cal} + 1, \dots, \tau_{end}$
- Rigid motion propagation

Global tool-tip location

$$\hat{\mathbf{p}}_B(t) = \mathbf{R}_t \hat{\mathbf{p}}_B(0) + \mathbf{d}_t$$